

CSCE 717 ALGORITHMIC GAME THEORY · MODULE 07 · LECTURE NOTES

Single-Item Auctions

Where the theory is sharpest, and most deployed.

MODULE OUTCOMES

- Analyse the four standard auction formats.
- Prove the second-price auction is truthful.
- State and explain revenue equivalence.
- Explain reserve prices and revenue maximisation.
- Explain why the theory's predictions sometimes fail.

1 The formats

Format	The rule	The optimal strategy
First price, sealed bid	Highest bid wins and pays its own bid.	Shade below your true value — by how much depends on your beliefs about the others, which makes it strategically demanding.
Second price, sealed bid (Vickrey)	Highest bid wins and pays the second-highest bid.	Bid your value. Dominant strategy — see the callout.
Dutch (descending clock)	The price falls until somebody accepts.	Strategically identical to first price — accepting at price p is the same decision as bidding p .
English (ascending)	The price rises until one bidder remains.	Stay in until the price reaches your value — which gives the second-price outcome.

Dutch is strategically identical to first price, and English to second price, in the independent private value setting — which is why there are two ideas here rather than four. Two formats, two strategies, and the pairing is the first result of auction theory; it also breaks when values are correlated (§4), which is informative about where the equivalence comes from.

IN A SECOND-PRICE AUCTION, BIDDING YOUR TRUE VALUE IS A DOMINANT STRATEGY

Your bid determines only whether you win, and never what you pay — the price is set entirely by somebody else — which is the whole mechanism, and the rest is two cases.

Consider bidding above your value: it changes the outcome only in the cases where you now win at a price above your value, which is a loss, and otherwise changes nothing at all.

And bidding below your value: it changes the outcome only in the cases where you now lose an auction you would have won profitably — also a loss, and otherwise nothing.

So truthful bidding weakly dominates every alternative — requiring no assumption whatsoever about the other bidders, their numbers, or their values — which is why this is the model mechanism of the entire field and is exactly Module 06 §3's VCG specialised to one item.

2 Revenue equivalence

UNDER ITS CONDITIONS, ALL FOUR FORMATS GIVE THE SELLER THE SAME EXPECTED REVENUE

Any two mechanisms that allocate the item to the same bidder (as a function of values) and give a bidder with the lowest possible value zero expected utility yield the same expected revenue — which is a genuinely remarkable theorem, since the formats look so different.

So the choice of format does not matter for revenue — under independent private values, risk-neutral bidders, symmetric value distributions, and no budget constraints, which is the list of hypotheses and is the part that matters.

And every one of those conditions fails somewhere: correlated values favour the English auction (the linkage principle), risk aversion favours first price, asymmetry breaks the symmetry assumption, and budget constraints change everything including who can win at all.

Which is why real auction design is about the conditions rather than about the formats — and why quoting revenue equivalence without its hypotheses is the characteristic error in this area, exactly as dropping the latency function class is in Module 04 §2.

3 Reserve prices

THE EFFICIENT AUCTION

always sell, to the highest bidder.
Maximises total value.

THE REVENUE-MAXIMISING AUCTION

set a reserve price, and do NOT sell below it.
Sometimes the item goes unsold even though a
bidder values it above the seller's cost --
which is INEFFICIENT ON PURPOSE.

WHY IT RAISES REVENUE

the reserve extracts more from the cases where
you do sell, at the cost of the cases where you
do not. With a single bidder it is the whole
mechanism: the reserve is the only thing
standing between the seller and a price of zero.

MYERSON'S RESULT

the optimal auction allocates by VIRTUAL value
rather than value -- a transformation of the bid
by the distribution it is drawn from.
Which means the optimal auction DEPENDS ON THE
DISTRIBUTION, and a seller who does not know it
cannot run it.

The optimal auction depends on the value distribution, so a seller who does not know that distribution cannot run it — which is why simple near-optimal auctions matter and is a substantial research area (a second-price auction with a well-chosen reserve gets most of the way, which is a satisfying result). Efficiency and revenue are different objectives, and the reserve price is where they visibly diverge — which is worth knowing before assuming an auction designer wants efficiency.

4 When it fails

- Bidders shade their bids in second-price auctions anyway — because they do not believe the rule, or do not understand it, or suspect the seller of shill bidding — which is measured in both laboratory and field settings and is a real limitation on 'provably truthful'.
- Repeated interaction changes everything — today's bid reveals information that affects tomorrow's price and tomorrow's competitors — which a single-shot analysis omits entirely, and which explains much of the shading above.
- Collusion is profitable and does happen — and second-price auctions are particularly vulnerable, since a bidding ring need only suppress one competing bid to lower the price, with no need to coordinate on anything else.
- Common values create the winner's curse: when all bidders are estimating the same unknown value, winning means you were the most optimistic estimator, so bidding your estimate loses money on

average. This is a selection effect rather than a mistake, and the correction is to bid below your estimate by an amount growing with the number of bidders.

- **And budget constraints break the model** in ways that cannot be patched, **which is why deployed advertising auctions depart substantially from the textbook versions** (CSCE 670's sponsored search material is the applied side of this).

SO THE HONEST SUMMARY OF THIS MODULE

The single-item theory is unusually complete — truthfulness, revenue equivalence, and the revenue-optimal auction are all settled results — which is rare in any applied field and is why this is where the subject is taught from.

And its predictions hold approximately, under conditions that are frequently violated in practice — which is the normal situation for an applied theory and is not a criticism of it, provided the conditions are stated.

So the practical skill is knowing which condition is binding in your case: correlated values, risk aversion, budget constraints, repetition, or collusion — five candidates, and usually exactly one of them explains the observed deviation, which makes it a tractable diagnosis.

Which is this module's version of the program's rule: name the auction, name the value model, and name the condition you are assuming — because 'the second-price auction is truthful' is a theorem with hypotheses, and the hypotheses are where the engineering is.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Vickrey — Counterspeculation, auctions, and competitive sealed tenders	§1 — the second-price auction in the original, and the argument is the one in the callout.
Myerson — Optimal auction design	§3 — virtual values and the revenue-optimal auction. Library copy.
Krishna — Auction Theory	§§2 and 4 — revenue equivalence derived properly, with every condition made explicit. Library copy.
Kagel & Levin — the experimental auction literature	§4 — what bidders actually do, measured, including the winner's curse in the field. Library copy.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Describe all four formats** and their optimal strategies.
- 2 **Show that Dutch and first price are strategically identical.**
- 3 **Prove second-price truthfulness**, both cases, in your own words.
- 4 **Simulate all four** with the same value draws and compare revenue.
- 5 **Violate one revenue equivalence condition** and watch the equivalence break.
- 6 **Compute the optimal reserve** for a uniform value distribution.
- 7 **Show that a reserve price is sometimes inefficient.**
- 8 **Simulate a common-value auction** and measure the winner's curse.
- 9 **Simulate a two-bidder ring** in a second-price auction.
- 10 **Find a deployed auction** and identify which condition it violates.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Name the four formats and their strategies.
- 2 Which pairs coincide, and under what assumption?
- 3 Prove second-price truthfulness.
- 4 Why does the bid not affect the price?
- 5 State revenue equivalence and its four conditions.
- 6 Give a case where each condition fails.
- 7 Why is the revenue-maximising auction inefficient?
- 8 What does Myerson's optimal auction depend on, and why does that matter?
- 9 Give five reasons the theory's predictions fail.
- 10 Explain the winner's curse as a selection effect.